

TMF8829 Optical Design Guide (ODG)

1v0

24 Oct 2025

TMF8829 Optical design guide

The design goal is to minimize system xtalk to stay below maximum levels – for TMF8829 unlike ams-OSRAM other dToF sensors, has no minimum requirement for xtalk as TMF8829 calibration is done ams-OSRAM factory.

The main factors for controlling system xtalk are airgap, glass thickness, glass apertures, optical barrier, and ink/tint characteristics.

General system recommendation are for dual modes – see TMF8829 datasheet section 7.3 and 7.3.1:

- Glass/TMF8829 airgap according to later slides; IR ink recommended for operation
- Optical barrier design (e.g. rubber boot) is recommended
- Glass ink with 85% or better IR transmissivity
- Assembly XY tolerance between glass apertures and TMF8829 is $\pm 0.20\text{mm}$ maximum

Notes

- For the non-dual modes, a much smaller crosstalk is recommended; otherwise the device minimum detection range is affected
- For best performance use a zero airgap design, where the TMF8829 is gently pushed to the cover glass

TMF8829 Field of view

The file *TMF8829_with_FOI_FOV.STEP* shows the

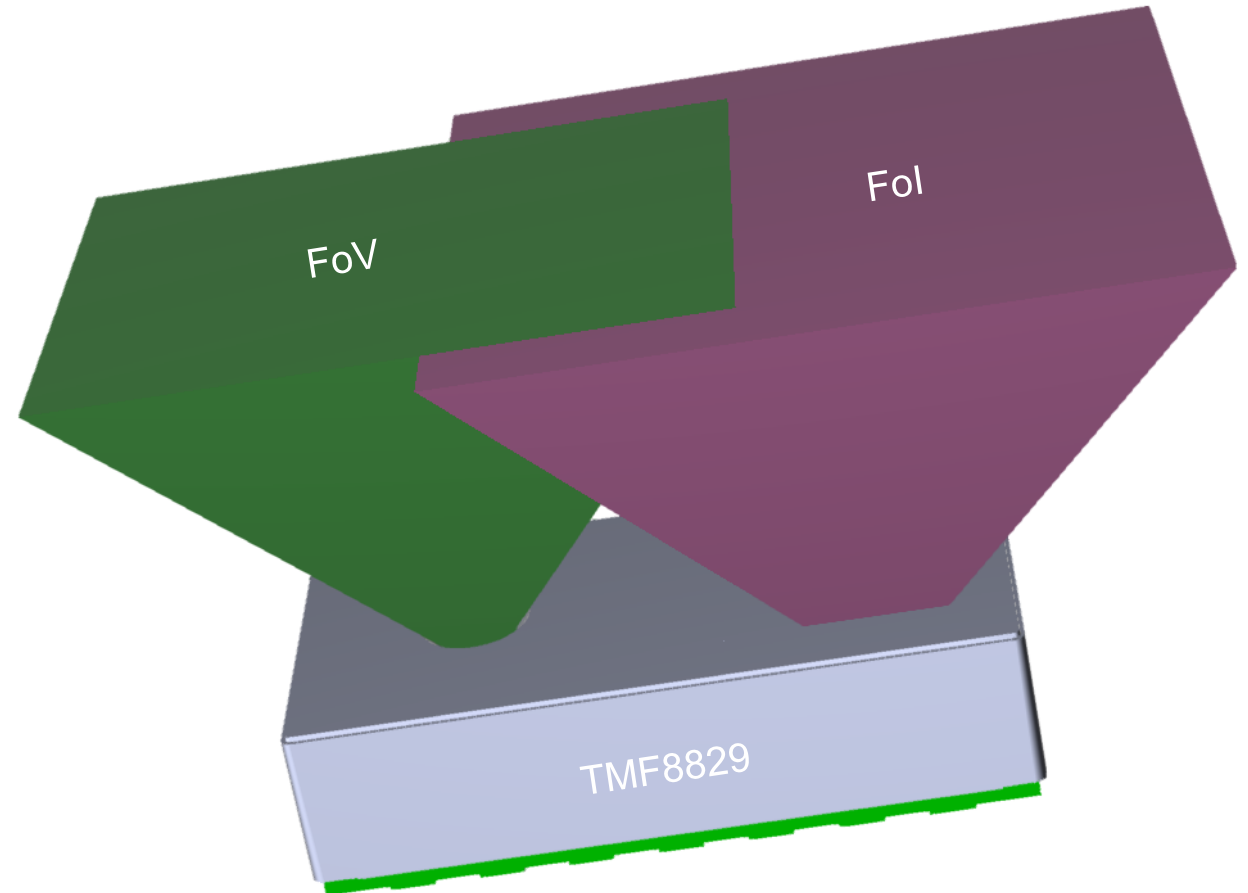
- TMF8829 device outline
- FoI – field of illumination of the VCSELs + MLA
- FoV – field of view of the receiver (lens + SPADs)

As STEP file for use in customer designs.

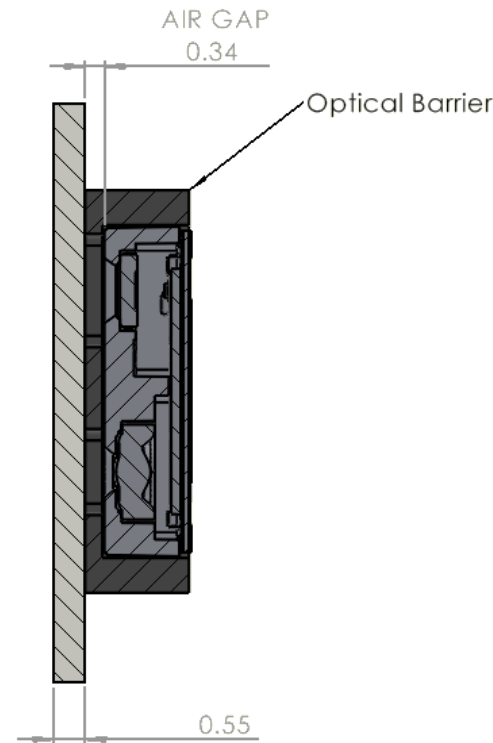
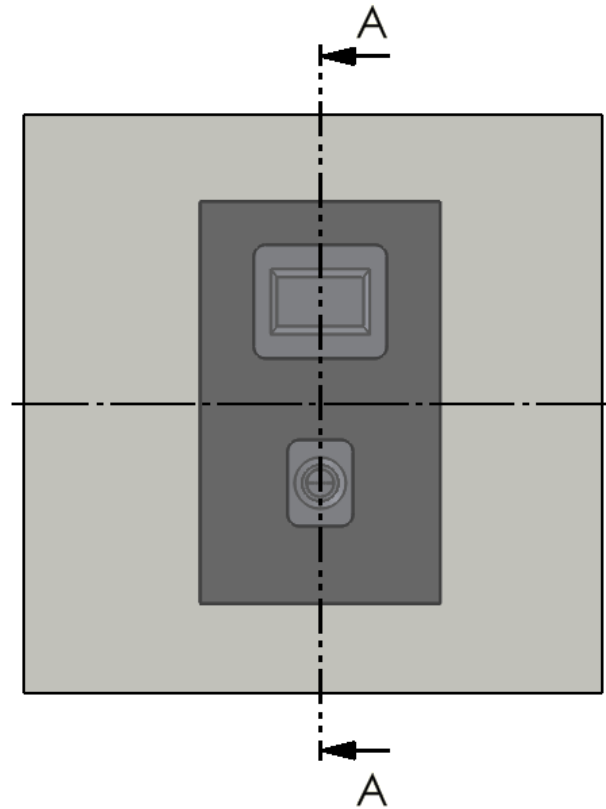
There is *TMF8829_with_FOI_FOV.pdf* with 3d content if a step viewer is not available.


TMF8829_with_FOI_FOV.STEP


Adobe Acrobat
Document



TMF8829 Optical Assembly Recommendation



SECTION A-A

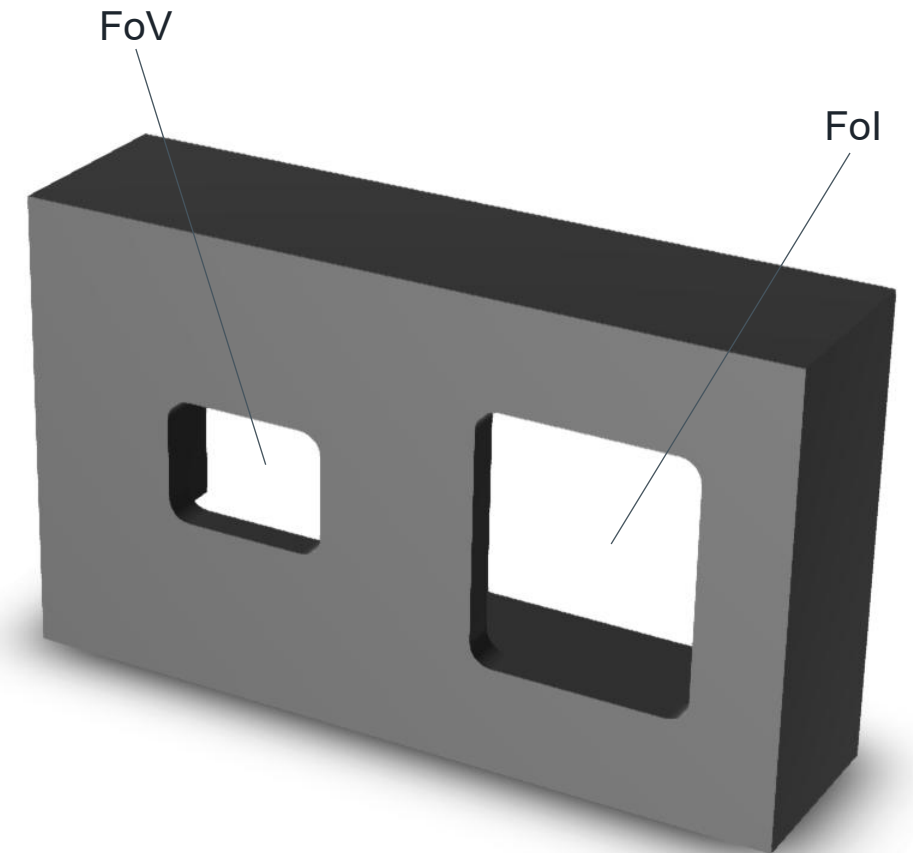
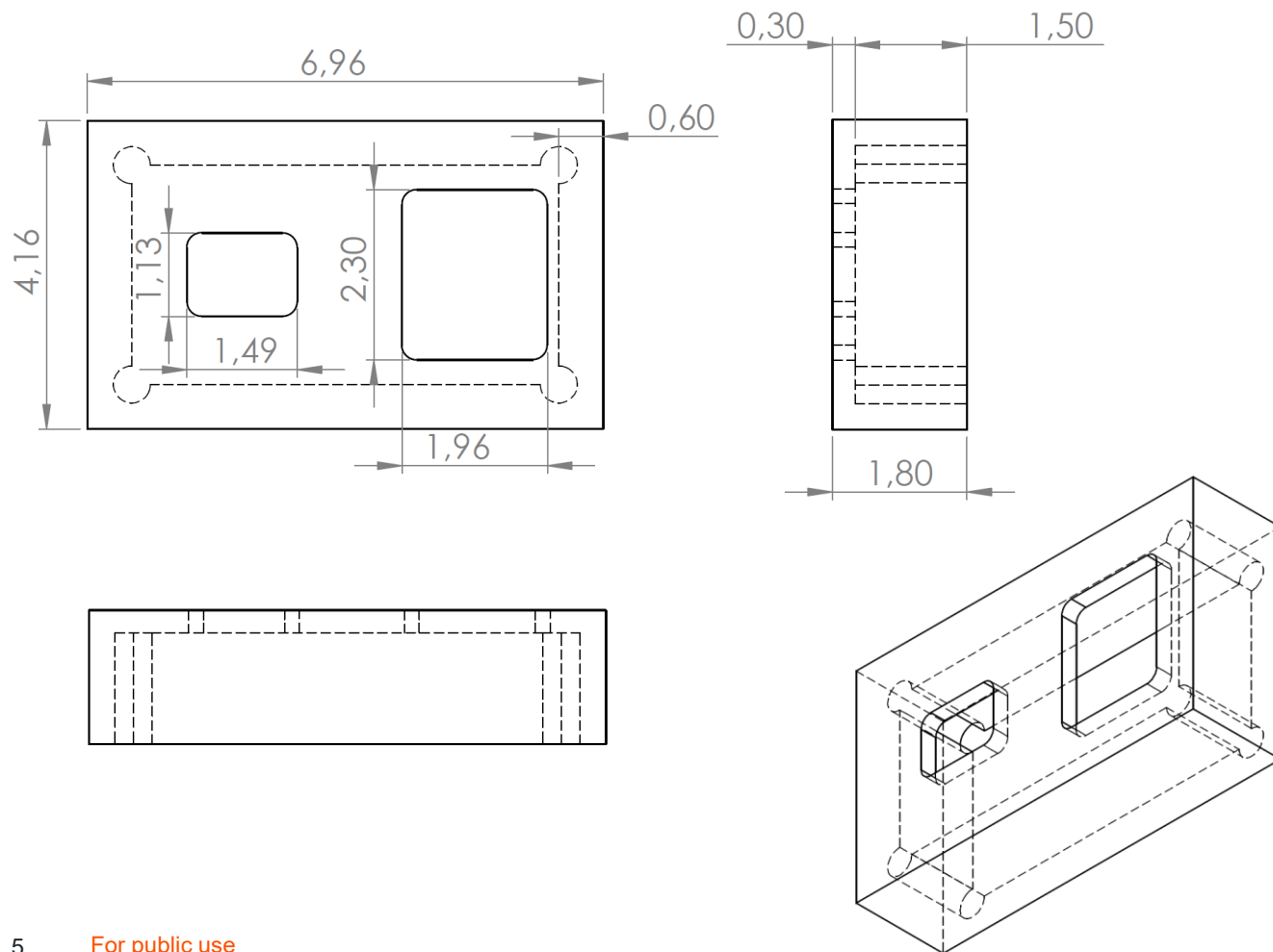
Material recommendations:

- **Isoprene (Rubber):** It is a special kind of polymer generally opaque to visible and infrared light.
- **Acrylic and Polycarbonate:** These are types of plastic optical components that can be used without problems at 940nm.
- **Silicon:** Silicon transmits IR but not visible light, making it opaque in the visible spectrum while still allowing IR transmission

Please note that the opacity can vary depending on the specific type of material and any additives used in its formulation. Therefore, ams OSRAM recommends to run material and optical characterizations before choosing a material.

TMF8829 Optical rubber boot design

See file *TMF8829_Rubber_Boot_0mm3.STEP*



TMF8829 Example cover glass

See file *TMF8829_Shield_Board_Hornix_Cover_Glass_RevA.pdf*



Adobe Acrobat
Document

This cover glass is used for

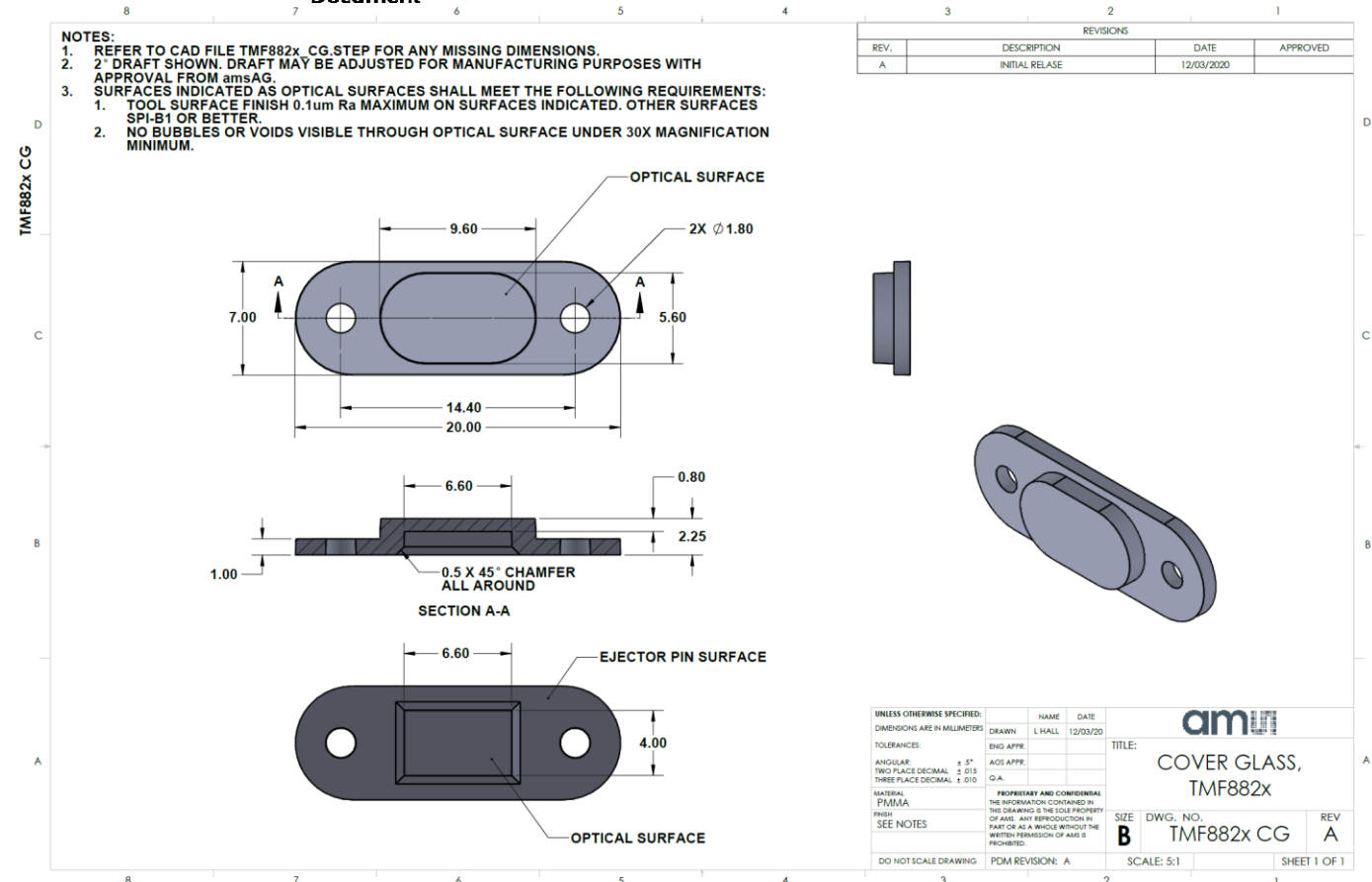
[TMF8829_EVM_EB_SHIELD Evaluation kit](#)

It can be ordered from

- <https://www.hornix.com.tw/en/home/>
Hornix Optical Technology Inc.
Address : No. 8, Qinghua 2nd St.,
Xinwu Dist., Taoyuan City 32742,
Taiwan (R.O.C.)
TEL : +886-3-4972665、886-3-4972180

Part name

- IR-T117-PM3D-A610
TMF882x_CG(0.8mm-2.25mm)



TMF8829 Maximum Crosstalk Values

Configuration for measurement

- Use 16x16 default mode configuration
- Have no target within 40cm
- No ambient IR light
- 600 k iterations

Value is peak weight of crosstalk peak -
use result 'xtalk' for each pixel

Calculate percentage of crosstalk per
sub-frame: For 600k iterations, max
<xtalk-value> / 300k * 100 [%]

Shown in the GUI

**For dual modes
crosstalk shall be < 7 %**



TMF8829 Optical design crosstalk measurements

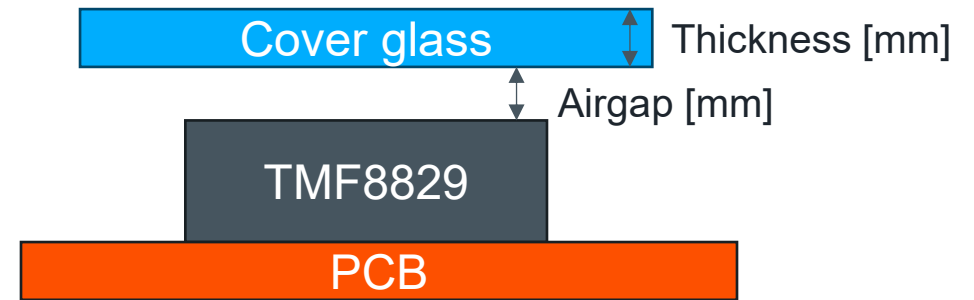
With / without rubber boot

To achieve maximum 7% crosstalk, see below measurements

Design without rubber boot Airgap [mm]	Glass thickness	
	0.55 mm	0.8 mm
0	0.09%	0.13%
0.1	0.22%	0.52%
0.2	0.38%	2.34%
0.3	1.50%	5.60%
0.4	4.25%	11.25%
0.5	9.82%	18.92%

Design with 0.3mm rubber boot Airgap [mm]	Glass thickness	
	0.55 mm	0.8 mm
0	Depends on compressibility of rubber boot	
0.1		
0.2		
0.3	0.37%	2.08%
0.4	1.49%	3.17%
0.5	3.78%	5.47%
0.6	7.33%	11.17%

Design without rubber boot



Design with rubber boot

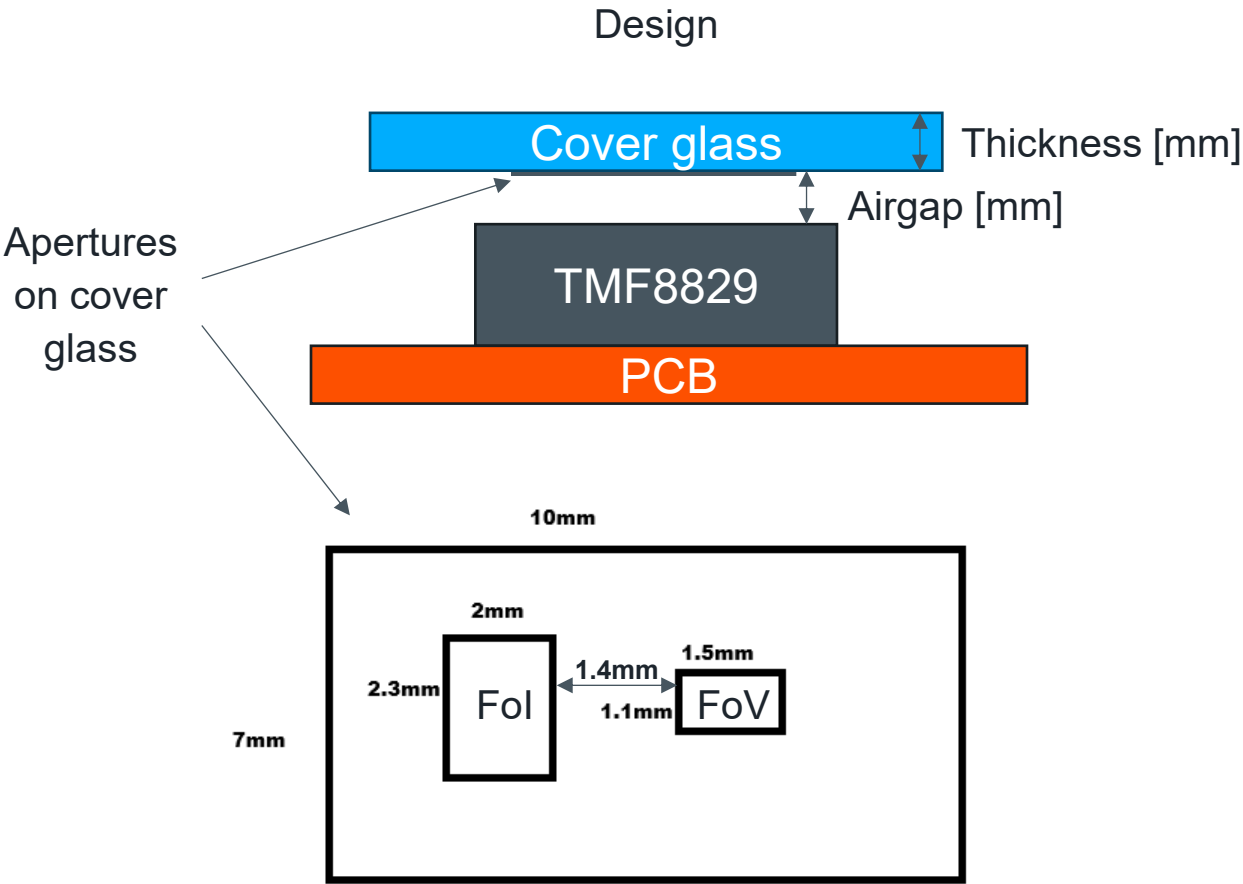


TMF8829 Optical design crosstalk measurements

With apertures on the cover glass

To achieve maximum 7% crosstalk, see below measurements

Design with apertures on cover glass	Glass thickness	
	Airgap [mm]	
Airgap [mm]	0.55 mm	0.8 mm
0	0.04%	0.12%
0.1	0.04%	0.15%
0.2	0.14%	0.21%
0.3	0.39%	0.37%
0.4	0.60%	0.72%
0.5	1.52%	1.41%
0.6	5.29%	3.45%
0.7	18.33%	7.74%

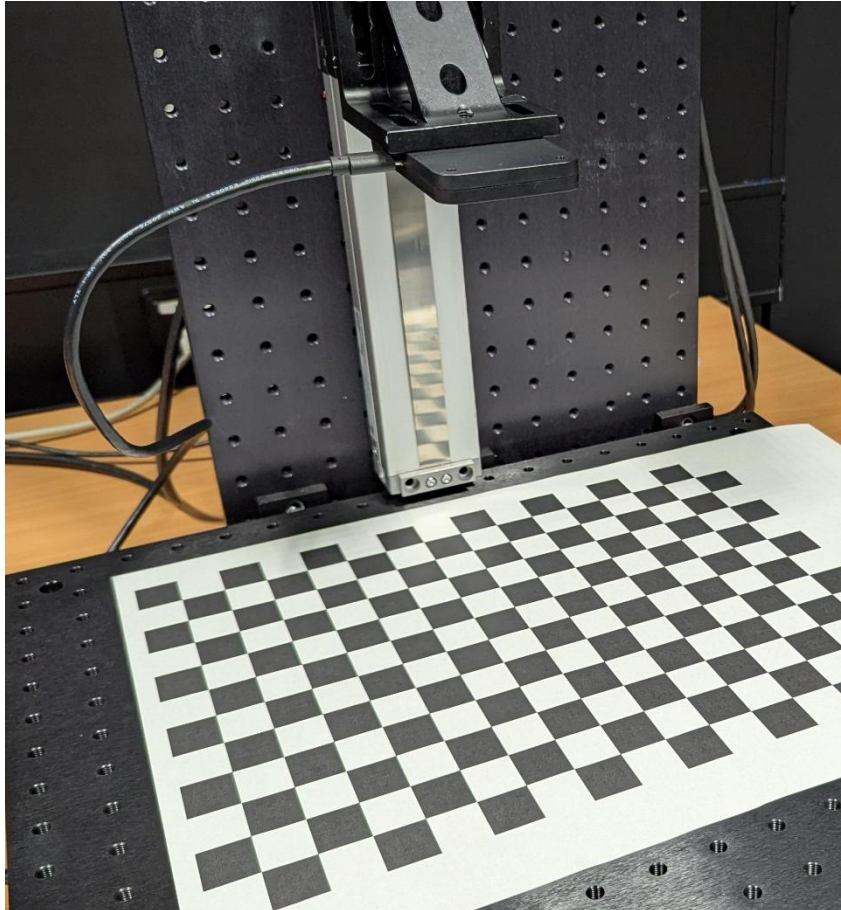


Notes:

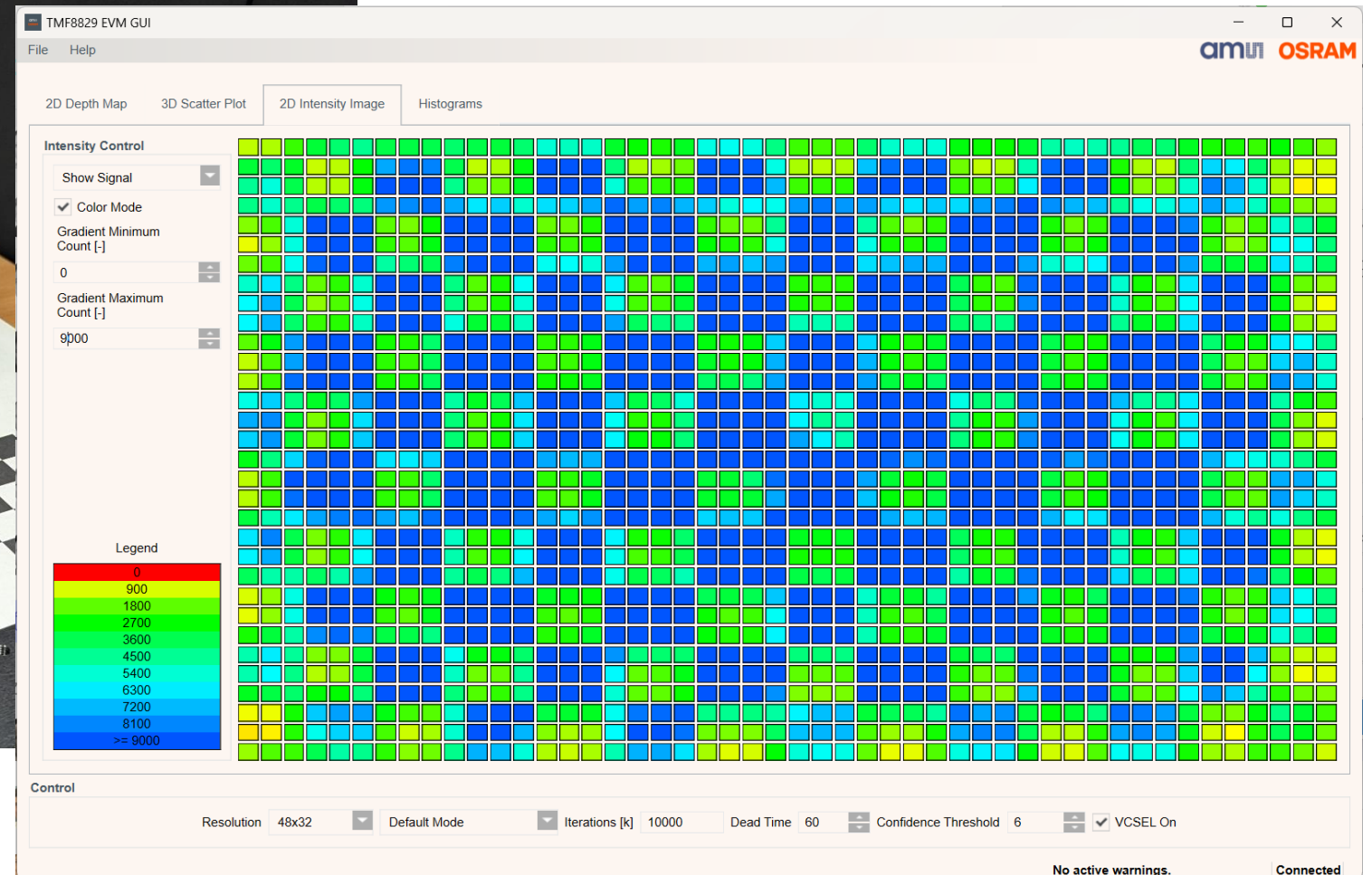
- Position FoI above VCSELs and FoV above receiver of TMF8829
- The apertures on cover glass are almost identical to *TMF8829_Rubber_Boot_0mm3.STEP* top part

TMF8829 Field of View (FOV) – for FoI see datasheet

FOV = 67.9 degrees x 52.8 degrees

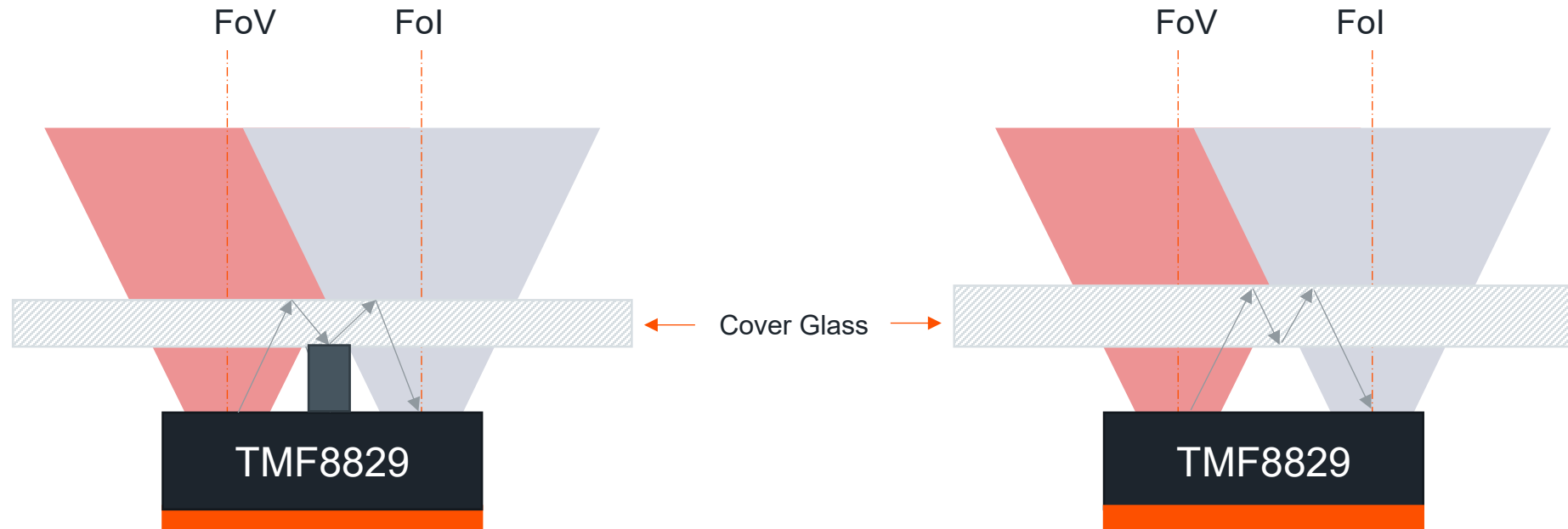


Checkerboard measurement result



TMF8829 Controlling crosstalk

Barrier, airgap and glass thickness



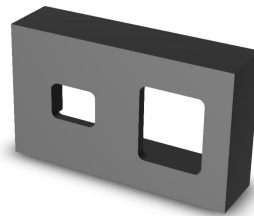
Crosstalk between device and cover glass
A larger airgap will increase crosstalk



Optical barrier to limit airgap crosstalk



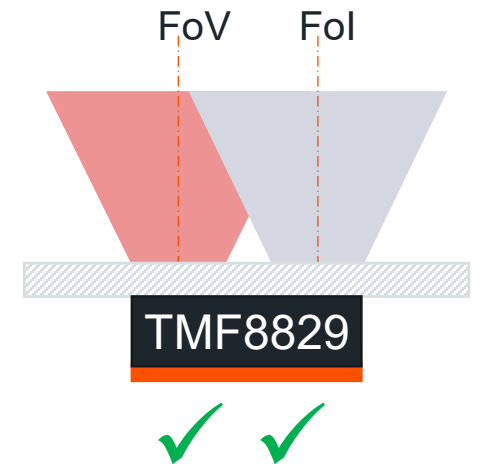
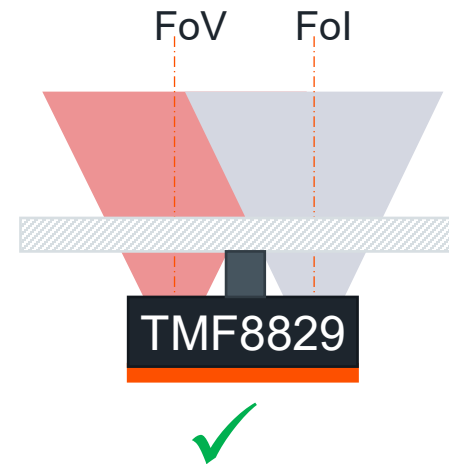
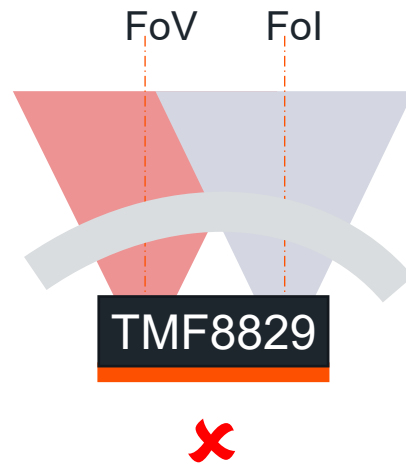
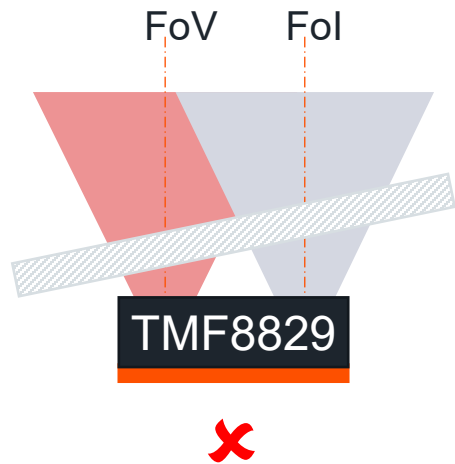
Crosstalk within cover glass
A thicker cover glass will increase crosstalk



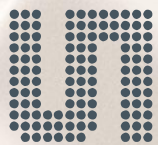
TMF8829 Controlling crosstalk

Glass design

- Keep airgap with limits defined by this document – minimize airgap, even go to zero airgap design
- Use optical isolation barrier (rubber boot) to reduce crosstalk
- Use cover glass with > 90% 940nm transmissivity
- Ensure crosstalk is within Optical Design Guide max range
- Keep cover glass perpendicular to FoI / FoV & no curved, textured or patterned glass



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